

Exercise sheet 5

Exercise 1 (1) Let $(X, \mathcal{B}, \mu)$ be a probability space, and let $f : X \rightarrow X$ be a measurable map that preserves μ . Assume f is bijective and its inverse is measurable. Let $\phi : X \rightarrow \mathbb{C}$ be a μ -summable function. For every x in X , we define

$$S_n \phi(x) = \frac{1}{n} \sum_{k=0}^{n-1} \phi(f^k(x)) \quad \text{and} \quad S_{n,-} \phi(x) = \frac{1}{n} \sum_{k=0}^{n-1} \phi(f^{-k}(x)).$$

Show that for μ -almost every x in X , the limits

$$\tilde{\phi}(x) = \lim_{n \rightarrow \infty} S_n \phi(x) \quad \text{and} \quad \tilde{\phi}_-(x) = \lim_{n \rightarrow \infty} S_{n,-} \phi(x)$$

exist and are equal.

(2) We now take $X = \mathbb{T}^N$, $\mathcal{B}$ the Borel σ -algebra, and $\mu = d\theta$ the Haar measure. Let $E = \mathbb{R}^N$, $p : E \rightarrow X$ the canonical projection, $\|\cdot\|$ a norm on E , and d the quotient distance on X , given by $d(\theta, \theta') = \inf\{\|v - v'\| \mid p(v) = \theta, p(v') = \theta'\}$. Let $\psi : X \rightarrow \mathbb{C}$ be a Lipschitz function with Lipschitz constant $Lip(\psi)$.

Show that for every θ in X and v in E , we have

$$|\psi(\theta + p(v)) - \psi(\theta)| \leq Lip(\psi) \|v\|.$$

(3) Let M be an $N \times N$ matrix with integer coefficients such that $\det M = \pm 1$, and let f_M be the transformation of X such that $p \circ M = f_M \circ p$.

1. Show that f_M is bijective. We henceforth take $f = f_M$.
2. For θ in X , let $\tau_\theta : X \rightarrow X$ be the translation by θ . Show that

$$f \circ \tau_\theta = \tau_{f(\theta)} \circ f.$$

(4) We now assume that M is hyperbolic. We choose for $\|\cdot\|$ a norm adapted to M , and we denote $E = E^s \oplus E^u$ the decomposition of E into M -invariant subspaces such that

$$ch(M) = \max\{\|M|_{E^s}\|, \|M^{-1}|_{E^u}\|\} < 1.$$

1. Show that for every θ in X and v in E^s , we have

$$|S_n \psi(\theta + p(v)) - S_n \psi(\theta)| \leq \frac{Lip(\psi)}{n(1 - ch(M))} \|v\|.$$

2. Show that for every v in E^s , we have $\tilde{\psi} \circ \tau_{p(v)} = \tilde{\psi}$ μ -almost everywhere.

3. Show that for every v in E^u , we have $\tilde{\psi} \circ \tau_{p(v)} = \tilde{\psi}$ μ -almost everywhere.
4. Show that for every v in E , we have $\tilde{\psi} \circ \tau_{p(v)} = \tilde{\psi}$ μ -almost everywhere.
5. Show that $\tilde{\psi}$ is constant μ -almost everywhere.
6. Show that $\tilde{\phi}$ is constant μ -almost everywhere.
7. Deduce that f is ergodic for μ .